Rahin Arefin Ahmed

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TECHNICAL SKILLS

- **Languages:** Python, C/C++, SQL, JavaScript.
- **Web Development:** Flask, HTML5, CSS3, Bootstrap 5, SQLAlchemy.
- **Data Science & AI:** Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, Computer Vision.
- **Tools & Platforms:** Git, GitHub, Jupyter Notebooks, PythonAnywhere.

EDUCATION

East West University Dhaka, Bangladesh
Bachelor of Science in Computer Science and Engineering 2022 – Present

PROJECTS

RokomariBG – Bangla Book Recommendation Graph Dataset *Python, Data Engineering*

- Built a large-scale, multi-entity heterogeneous knowledge graph dataset for Bangla book recommendation research.
- The dataset includes 127K+ books, 63K+ users, 16K+ authors, 1.5K+ categories, and 209K+ reviews connected through multiple relational types.
- Designed to support research on recommendation systems in low-resource languages and enable benchmarking of collaborative filtering, graph neural networks, and retrieval models.
- **GitHub:** github.com/backlashblitz/Bangla-Book-Recommendation-Dataset

Object Detection with SSL & Semi-Supervised Learning *Python, Computer Vision, YOLO*

- Developed an object detection pipeline using Self-Supervised Learning (SSL) and Semi-Supervised Learning with YOLOv10–YOLOv12 architectures.
- Explored techniques for improving detection performance with limited labeled data by leveraging large unlabeled datasets.
- Implemented training workflows, dataset preprocessing, and model evaluation for robust object detection experiments.
- **GitHub:** github.com/backlashblitz/FishFryVision-YOLOv10-to-v12

RESEARCH PUBLICATIONS

Towards Personalized Bangla Book Recommendation: A Large-Scale Multi-Entity Book Graph Dataset
arXiv Preprint (cs.IR, cs.LG)

- Introduced **RokomariBG**, a large-scale heterogeneous knowledge graph dataset for Bangla book recommendation research.
- **arXiv:** arxiv.org/abs/2602.12129 Feb 2026

Explainable Machine-Learning Forecasts of Building-Energy Demand from Weather Signals
2025 International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN)

- Conducted a comparative study of Classical, Ensemble and Hybrid DL Models.
- **DOI:** 10.1109/QPAIN66474.2025.11172015 July 2025

CERTIFICATIONS & AWARDS

- **Galactic Problem Solver** – NASA International Space Apps Challenge 2025
  Awarded for outstanding participation and contributions in addressing global challenges affecting Earth and space during the NASA Space Apps Challenge.
- **Supervised Machine Learning: Regression and Classification** – DeepLearning.AI & Stanford Online
  Completed intensive specialization on building and evaluating machine learning models.
- **Certificate of Presentation (IEEE QPAIN 2025)** – IEEE Photonics Society Bangladesh Chapter
  Successfully presented research paper at QPAIN 2025.
- **Graphic Design Masterclass Live Course** – eShikhon
  Completed comprehensive training, achieving 90% marks over a 4+ month duration.